

The Aperioc mechanism: attention through latent discovery

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Abstract

Deep learning the hidden key to implicit attention.
Reveal latent meaning, focus on what matters.

Introduction

An implicit deep learning model [2] consists of a pair of equations, an equilibrium equation E and a prediction equation P. E is function of the input and a hidden state, in equilibrium it passes a fixed point state to P, a function of this fixed point and the input. E and P are each parameterized by pair of weight matrices to be learned.

The **Aperioc*** mechanism, attention via an implicit model:

$$(E) \quad s_i = \varphi(K s_i + Q t_i)$$

$$(P) \quad 0 = V s_i + Q' t_i$$

where t_i is an input token, φ is a nonlinear activation function, K , Q , V and Q' are weight matrices to be learned, and s_i is a fixed point (an implicit source token).

Interpret E and P using:

query $q_i = Q t_i$

key $k_i = K s_i$

value $v_i = V s_i$

updated query $q_i' = - Q' t_i$

Sufficient conditions [3] have been established for the key weight matrix K to be well posed for the activation function φ , that is (E) has a unique fixed point solution.

The key-value pairs embody value meaning. Attention mechanisms [1] create a contextual representation z_i of target t_i , using a probability weighted sum of values, the weights reflecting the similarity between updated queries and keys. Note that learning a cascade of Aperioc mechanisms (implicit models) is a structured way to embed tokens and update queries, and thereby update contextual representations; yielding a coherent stackable attention mechanism. Updates are performed when the average Renyi α -entropy of the probability weights is lowered.

* “aperio occultus” Reveal the hidden.

Transformer key-query-value self-attention [1]

Input tokens $t_1, \dots, t_m \in \mathbb{R}^r$

Weight matrices $Q, K, V \in \mathbb{R}^{n \times r}$ define:

$$\text{query } q_i \triangleq Q t_i$$

$$\text{key } k_i \triangleq K t_i$$

$$\text{value } v_i \triangleq V t_i$$

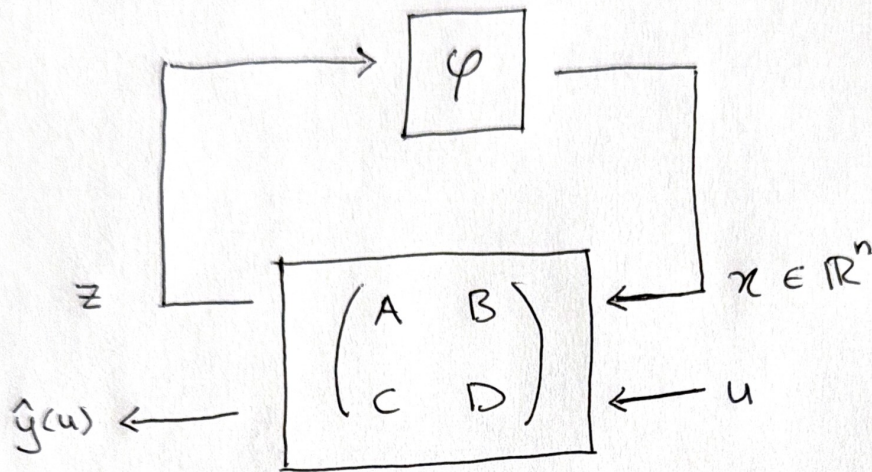
For $i \in [m]$, let $\alpha_{ij} \triangleq \text{softmax}_j \langle q_i, k_j \rangle$

Contextual representation z_i of t_i

$$z_i \triangleq \sum_{j=1}^m \alpha_{ij} v_j$$

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Implicit deep learning [2]



For a given input vector u , the predicted vector is

$$\hat{y}(u) = Cx + Du \quad (\text{prediction equation})$$

$$x = \varphi(Ax + Bu) \quad (\text{equilibrium equation})$$

where $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a nonlinear activation function

and matrices A, B, C, D are parameters to be learned.

Think of x as a "state" corresponding to "hidden"

features that are extracted from the inputs, based on the so-called equilibrium equation.

Training implicit models [3]

Given $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times r}$, $C \in \mathbb{R}^{q \times n}$, $D \in \mathbb{R}^{q \times r}$

consider the implicit model

$$x = \varphi(Ax + Bu), \quad y = Cx + Du \quad (+)$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^r$, $y \in \mathbb{R}^q$

and $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined by $\varphi(x) = (\varphi_1(x_1), \dots, \varphi_n(x_n))$.

For $i \in [n]$, assume the activation function $\varphi_i: \mathbb{R} \rightarrow \mathbb{R}$

is weakly increasing i.e. $\varphi_i(x_i) \geq \varphi_i(z_i)$ for $x_i \geq z_i$,

and non-expansive i.e. $|\varphi_i(x_i) - \varphi_i(z_i)| \leq |x_i - z_i|$

for all x_i and z_i ; if φ_i is differentiable, these

conditions are equivalent to $0 \leq \varphi_i'(x_i) \leq 1$

for all $x_i \in \mathbb{R}$.

Given an input data matrix $U = [u_1, \dots, u_m] \in \mathbb{R}^{r \times m}$

and a corresponding output matrix $Y = [y_1, \dots, y_m] \in \mathbb{R}^{q \times m}$,

we aim to learn matrices A, B, C, D so that

(†) approximates the input-output relationship.

Rewrite (†) for matrix inputs as:

$$\hat{Y} = CX + DU, \text{ where } X = \varphi(AX + BU).$$

If each φ_i is weakly increasing and non-expansive,

the fixed point problem is well-posed [3]

when $\mu_{\infty, [\eta]^{-1}}(A) < 1$, for some $\eta \in \mathbb{R}_{>0}^n$.

Given $\eta \in \mathbb{R}_{>0}^n$, let $[\eta]$ denote the diagonal matrix

with diagonal entries η . Matrix measure

$$\mu_{\infty, [\eta]^{-1}}(A) \triangleq \max_{i \in [n]} \left(a_{ii} + \sum_{j=1, j \neq i}^n |a_{ij}| \frac{\eta_j}{\eta_i} \right)$$

We learn A, B, C, D by solving the optimization problem:

$$\min_{A, B, C, D, X} \mathcal{L}(Y, CX + DU) + \mathcal{P}(A, B, C, D)$$

$$X = \varphi(AX + BU), \quad \mu_{\infty}[\varphi]^{-1}(A) \leq \gamma$$

where $\mathcal{L}$ is a loss function, $\mathcal{P}$ is a penalty function,

and $\gamma < 1$ is a hyperparameter ensuring the fixed point problem is well-posed. For details see [3].

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The Aperiodic mechanism

Recall the generic implicit model setup:

input $u_1, \dots, u_m \in \mathbb{R}^r$

output $y_1, \dots, y_m \in \mathbb{R}^q$

nonlinear activation function $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$(E) \quad \underset{n \times 1}{x} = \varphi \left(\underset{n \times n}{A} \underset{n \times 1}{x} + \underset{n \times r}{B} \underset{r \times 1}{u} \right)$$

$$(P) \quad \underset{q \times 1}{y} = \underset{q \times n}{C} \underset{n \times 1}{x} + \underset{q \times r}{D} \underset{r \times 1}{u}$$

A, B, C, D are matrices to be learned

(assuming A is well-posed for φ).

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Let $q = n$.

Input tokens $t_1, \dots, t_m \in \mathbb{R}^r$

Learn weight matrices K, Q, V, Q' so that:

Aperiodic
mechanism

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$$s_i = \varphi \left(\underset{n \times n}{K} s_i + \underset{n \times r}{Q} \underset{r \times 1}{t_i} \right)$$

$$\underset{n \times 1}{0} = \underset{n \times n}{V} s_i + \underset{n \times r}{Q'} t_i$$

Let query $q_i \triangleq Q t_i$

key $k_i \triangleq K s_i$

value $v_i \triangleq V s_i$

updated query $q'_i \triangleq -Q' t_i$

all vectors in $\mathbb{R}^n$, $i \in [m]$.

Fixed point s_i can be viewed as an implicit source token. We now give a contextual representation z_i of target t_i .

Let score be a similarity function between queries and keys, like a reproducing kernel, for example.

$$\text{Let } \alpha_{ij} \triangleq \underset{j \in [m]}{\text{softmax}} \text{score}(q'_i, k_j)$$

$$\text{and } j_i \triangleq \arg \max_j \alpha_{ij}.$$

In response to query q_i , return:

i) probability α_{ij_i}

ii) value v_{j_i}

$$\begin{aligned} \text{iii) contextual representation } z_i &\triangleq \sum_{j=1}^m \alpha_{ij} v_j \\ &= \sum_{j=1}^m \alpha_{ij} q'_j(t_j) \end{aligned}$$

Summary:

Via an implicit model, the Aperiodic mechanism learns embeddings K, Q, V, Q' and source tokens s_i for a key-value pairing, to update queries and generate a contextual representation z_i of target t_i , an implicit form of attention.

Contextual representation z_i updating:

Recall Renyi α -entropy.

For $0 < \alpha < \infty$, $\alpha \neq 1$,

$$H_\alpha(\{p_i\}_{i=1}^n) \triangleq \frac{1}{1-\alpha} \ln\left(\sum_{i=1}^n p_i^\alpha\right),$$

where $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$.

For $\alpha = 0, 1, \infty$ define $H_\alpha \triangleq \lim_{\gamma \rightarrow \alpha} H_\gamma$.

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For $i \in [m]$, define $\beta'_{ij} \triangleq \text{softmax score}(q'_i, k_j)_{j \in [m]}$.

$$\text{Define } E(\{q'_i\}_{i=1}^m) = \frac{1}{m} \sum_{i=1}^m H_\alpha^i$$

where $H_\alpha^i = H_\alpha(\{\beta'_{ij}\}_{j=1}^m)$.

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t_i

$$s_i = \varphi \left(\underset{\underset{\parallel}{K_i}}{K} s_i + \underset{\underset{\parallel}{q_i}}{Q} t_i \right)$$

represent
↓

$$O = \underset{\underset{\parallel}{V_i}}{V} s_i + \underset{\underset{\parallel}{q'_i}}{Q'} t_i$$

 z_i

$$z_i \triangleq \sum_{j=1}^m \alpha_{ij} V_j, \quad \alpha_{ij} [\text{score}(q'_i, k_j)]$$

update
↓

$$^1 s_i = \varphi \left(\underset{\underset{\parallel}{K_i}}{^1 K} ^1 s_i + \underset{\underset{\parallel}{q_i}}{^1 Q} z_i \right)$$

$$O = \underset{\underset{\parallel}{V_i}}{^1 V} ^1 s_i + \underset{\underset{\parallel}{q'_i}}{^1 Q'} z_i$$

 $^1 z_i$

$$^1 z_i \triangleq \sum_{j=1}^m ^1 \alpha_{ij} ^1 V_j, \quad ^1 \alpha_{ij} [\text{score}(^1 q'_i, ^1 k_j)]$$

update
↓

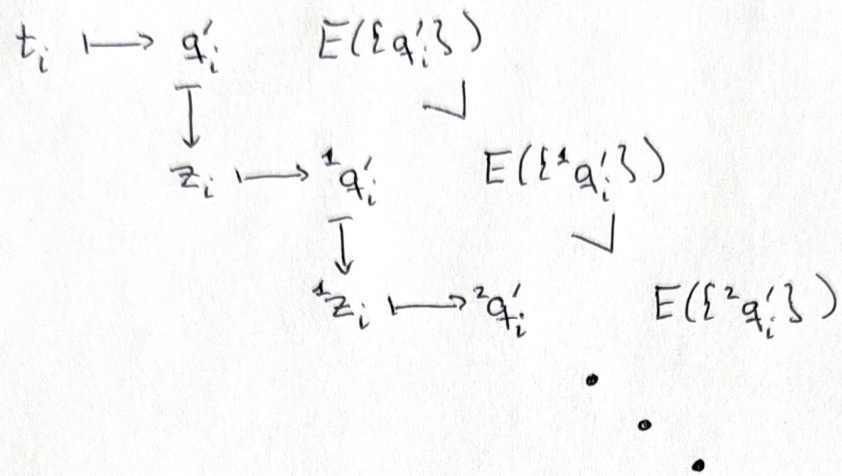
$$^2 s_i = \varphi \left(\underset{\underset{\parallel}{K_i}}{^2 K} ^2 s_i + \underset{\underset{\parallel}{q_i}}{^2 Q} ^1 z_i \right)$$

$$O = \underset{\underset{\parallel}{V_i}}{^2 V} ^2 s_i + \underset{\underset{\parallel}{q'_i}}{^2 Q'} ^1 z_i$$

 $^2 z_i$

$$^2 z_i \triangleq \sum_{j=1}^m ^2 \alpha_{ij} ^2 V_j, \quad ^2 \alpha_{ij} [\text{score}(^2 q'_i, ^2 k_j)]$$

and so on ...



Cascade implicit models so long as Renyi α -entropy is being lowered. This is how the Aperiodic mechanism updates contextual representations.

□

References

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